

Checking Optimality in Unconstrained Optimization

1 The Goal: Finding Local Minima (and Maxima)

We want to find points $\mathbf{x}^*$ where a multivariate function $f(\mathbf{x})$ reaches a local minimum. For an unconstrained problem, this means finding the “bottom of a valley” in the function’s landscape.

2 The Necessary First-Order Condition (FOC)

- **Theorem:** If f is continuously differentiable and $\mathbf{x}^*$ is a **local minimizer** (or maximizer), then the gradient of f at $\mathbf{x}^*$ must be the zero vector.

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

- A point where $\nabla f(\mathbf{x}) = \mathbf{0}$ is called a **critical point** or **stationary point**.
- This condition is **necessary but not sufficient**. A critical point can be:
 - A **local minimum**
 - A **local maximum**
 - A **saddle point**

How to check:

1. Compute the gradient vector $\nabla f(\mathbf{x})$.
2. Set each partial derivative in the gradient to zero.
3. Solve the resulting system of equations to find all critical points $\mathbf{x}^*$.

Example 1 (2D):

Find the critical points of $f(x, y) = x^2 + 2y^2 - 4x - 8y + 15$.

1. Compute the gradient:

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x - 4 \\ 4y - 8 \end{bmatrix}$$

2. Set the gradient to zero:

$$\begin{cases} 2x - 4 = 0 \\ 4y - 8 = 0 \end{cases}$$

3. Solve the system:

$$\begin{cases} x = 2 \\ y = 2 \end{cases}$$

The only critical point is $\mathbf{x}^* = (2, 2)$.

3 The Sufficient Second-Order Condition (SOC)

- **Theorem:** Suppose f is twice continuously differentiable and $\nabla f(\mathbf{x}^*) = \mathbf{0}$. Let $\mathbf{H}(\mathbf{x}^*)$ be the Hessian matrix at that point.
 - If $\mathbf{H}(\mathbf{x}^*)$ is **Positive Definite** $\Rightarrow \mathbf{x}^*$ is a **local minimizer**
 - If $\mathbf{H}(\mathbf{x}^*)$ is **Negative Definite** $\Rightarrow \mathbf{x}^*$ is a **local maximizer**
 - If $\mathbf{H}(\mathbf{x}^*)$ is **Indefinite** $\Rightarrow \mathbf{x}^*$ is a **saddle point**
 - If $\mathbf{H}(\mathbf{x}^*)$ is semi-definite, the test is **inconclusive**

How to check definiteness for a 2×2 Hessian:

For $\mathbf{H} = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$:

- **Positive Definite** if: $a > 0$ and $\det(\mathbf{H}) = ac - b^2 > 0$
- **Negative Definite** if: $a < 0$ and $\det(\mathbf{H}) = ac - b^2 > 0$
- **Indefinite** if: $\det(\mathbf{H}) = ac - b^2 < 0$

How to check for a 3×3 Hessian:

Check the signs of the **leading principal minors** (D_1, D_2, D_3). For $\mathbf{H} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$:

$$D_1 = a$$

$$D_2 = \det \begin{bmatrix} a & b \\ d & e \end{bmatrix} = ae - bd$$

$$D_3 = \det(\mathbf{H})$$

- **Positive Definite** if: $D_1 > 0, D_2 > 0, D_3 > 0$
- **Negative Definite** if: $D_1 < 0, D_2 > 0, D_3 < 0$

4 Applied Examples

Example 1 (2D) Continued:

Classify the critical point $(2, 2)$ for $f(x, y) = x^2 + 2y^2 - 4x - 8y + 15$.

1. Compute the Hessian Matrix:

$$\mathbf{H}(x, y) = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$$

2. Evaluate at $(2, 2)$: $\mathbf{H}(2, 2) = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}$

3. Check for definiteness:

- $a = 2 > 0$
- $\det(\mathbf{H}) = (2)(4) - (0)(0) = 8 > 0$
- **YES** The Hessian is **Positive Definite**

Conclusion: $(2, 2)$ is a **local minimizer**.

Example 2 (3D):

Find and classify all critical points of $f(x, y, z) = x^2 + y^2 + z^2 + xy - 4z$.

1. **FOC:**

- Gradient:

$$\nabla f = \begin{bmatrix} 2x + y \\ 2y + x \\ 2z - 4 \end{bmatrix}$$

- Set to zero:

$$\begin{cases} 2x + y = 0 \\ x + 2y = 0 \\ 2z - 4 = 0 \end{cases}$$

- Solve:

$$z = 2$$

$$y = -2x$$

$$x + 2(-2x) = 0 \Rightarrow -3x = 0 \Rightarrow x = 0$$

$$y = 0$$

Critical point: $\mathbf{x}^* = (0, 0, 2)$

2. SOC:

- Hessian Matrix:

$$\mathbf{H} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- Leading principal minors:

$$D_1 = 2 > 0$$

$$D_2 = \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 4 - 1 = 3 > 0$$

$$D_3 = \det \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2 \cdot \det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 2 \cdot 3 = 6 > 0$$

- **YES** All $D_i > 0$, so **Positive Definite**

Conclusion: $(0, 0, 2)$ is a **local minimizer**.

Example 3 (Saddle Point):

$$f(x, y) = x^2 - y^2$$

1. FOC:

$$\nabla f = \begin{bmatrix} 2x \\ -2y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (x, y) = (0, 0)$$

2. SOC:

$$\mathbf{H} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

- $a = 2 > 0$
- $\det(\mathbf{H}) = (2)(-2) - (0)(0) = -4 < 0$
- **NO** The Hessian is **Indefinite**

Conclusion: $(0, 0)$ is a **saddle point**.

Summary of Procedure

1. **Find Critical Points:** Solve $\nabla f(\mathbf{x}) = \mathbf{0}$
2. **Compute Hessian:** Find $\mathbf{H}(\mathbf{x})$
3. **Classify Each Critical Point:**
 - Evaluate $\mathbf{H}(\mathbf{x}^*)$

- Check definiteness:
 - **Positive Definite** $\Rightarrow$ Local Minimum **YES**
 - **Negative Definite** $\Rightarrow$ Local Maximum
 - **Indefinite** $\Rightarrow$ Saddle Point
 - **Semi-Definite** $\Rightarrow$ Test Inconclusive